

Manufacturing Knowledge Graph

GraphML to PDF visualization report

Source file: phase9_manufacturing_knowledge_graph.graphml

Generated: 22 June 2026

This PDF converts the uploaded GraphML into a readable manufacturing knowledge graph report. The full overview keeps feature labels hidden because the graph contains thousands of feature nodes. The line detail pages show the top critical features per station.

Metric	Value
Nodes	4,321
Edges	4,923
Directed graph	Yes
Node types	failure: 1, line: 4, station: 52, feature: 4,264
Relationship types	CONTAINS_STATION: 52, HAS_FEATURE: 4,264, TRANSITIONS_TO: 531, MODEL_ASSOCIATED_WITH_FAILURE: 76

Visual element	Meaning
Blue squares	Manufacturing lines
Orange squares	Stations
Gray dots/circles	Features
Red diamond	Failure target: Response=1
Red stations/arrows	Highest aggregated association with failure
Gray arrows/lines	Manufacturing transitions and feature links

Top critical lines

label	critical_score	evidence_value
L1	60.08	5.338e+05
L3	52.95	1.4683e+07
L0	39.60	1.2839e+07
L2	37.59	7.1378e+05

Top critical stations

label	line	critical_score	evidence_value
L1_S24	L1	60.08	3.6658e+05
L3_S29	L3	52.95	2.2391e+06
L3_S32	L3	50.90	48678
L3_S30	L3	47.34	2.2394e+06
L3_S34	L3	46.02	2.2301e+06
L3_S33	L3	44.63	2.2292e+06
L3_S39	L3	44.24	1.2002e+05
L1_S25	L1	42.76	1.6722e+05
L3_S37	L3	42.58	2.2407e+06
L0_S13	L0	39.60	4.8448e+05
L3_S38	L3	38.52	54175
L2_S26	L2	37.59	4.5434e+05
L2_S27	L2	37.36	2.4028e+05
L3_S35	L3	36.78	1.1042e+06
L3_S36	L3	35.46	1.1379e+06

Stations with highest aggregated failure evidence

station	failure_evidence
L1_S24	0.26177
L3_S32	0.068044
L2_S27	0.023317
L1_S25	0.022733
L2_S28	0.0073748
L3_S49	0.0072141
L3_S38	0.0015426
L0_S10	0.0014045

Top critical features

label	line	station	critical_score	evidence_value
L3_S32_F3850	L3	L3_S32	100.00	0.067988
L2_S27_F3144	L2	L2_S27	34.30	0.023317
L1_S24_F1844	L1	L1_S24	30.73	0.020889
L1_S24_F1846	L1	L1_S24	26.25	0.017843
L1_S24_F1723	L1	L1_S24	20.29	0.013797
L1_S24_F1778	L1	L1_S24	18.49	0.012571
L1_S24_F1667	L1	L1_S24	17.18	0.011678
L1_S24_F1498	L1	L1_S24	15.31	0.010412
L1_S24_F1713	L1	L1_S24	13.27	0.0090237
L1_S24_F1842	L1	L1_S24	12.42	0.0084438
L1_S24_F1758	L1	L1_S24	12.25	0.0083253
L1_S24_F958	L1	L1_S24	11.82	0.0080392
L1_S24_F1672	L1	L1_S24	11.05	0.0075121
L1_S24_F1609	L1	L1_S24	10.90	0.0074093
L1_S24_F1122	L1	L1_S24	10.75	0.0073106
L3_S49_F4211	L3	L3_S49	9.96	0.0067721
L1_S24_F1490	L1	L1_S24	9.95	0.0067678
L1_S24_F1002	L1	L1_S24	9.92	0.0067471
L1_S24_F1176	L1	L1_S24	9.63	0.0065496
L1_S25_F2990	L1	L1_S25	8.46	0.0057489
L1_S24_F1632	L1	L1_S24	8.37	0.0056924
L1_S24_F1174	L1	L1_S24	8.08	0.0054913
L1_S24_F1718	L1	L1_S24	7.59	0.005157
L1_S24_F963	L1	L1_S24	7.55	0.00513
L1_S24_F1695	L1	L1_S24	7.38	0.005015

Features most directly associated with failure

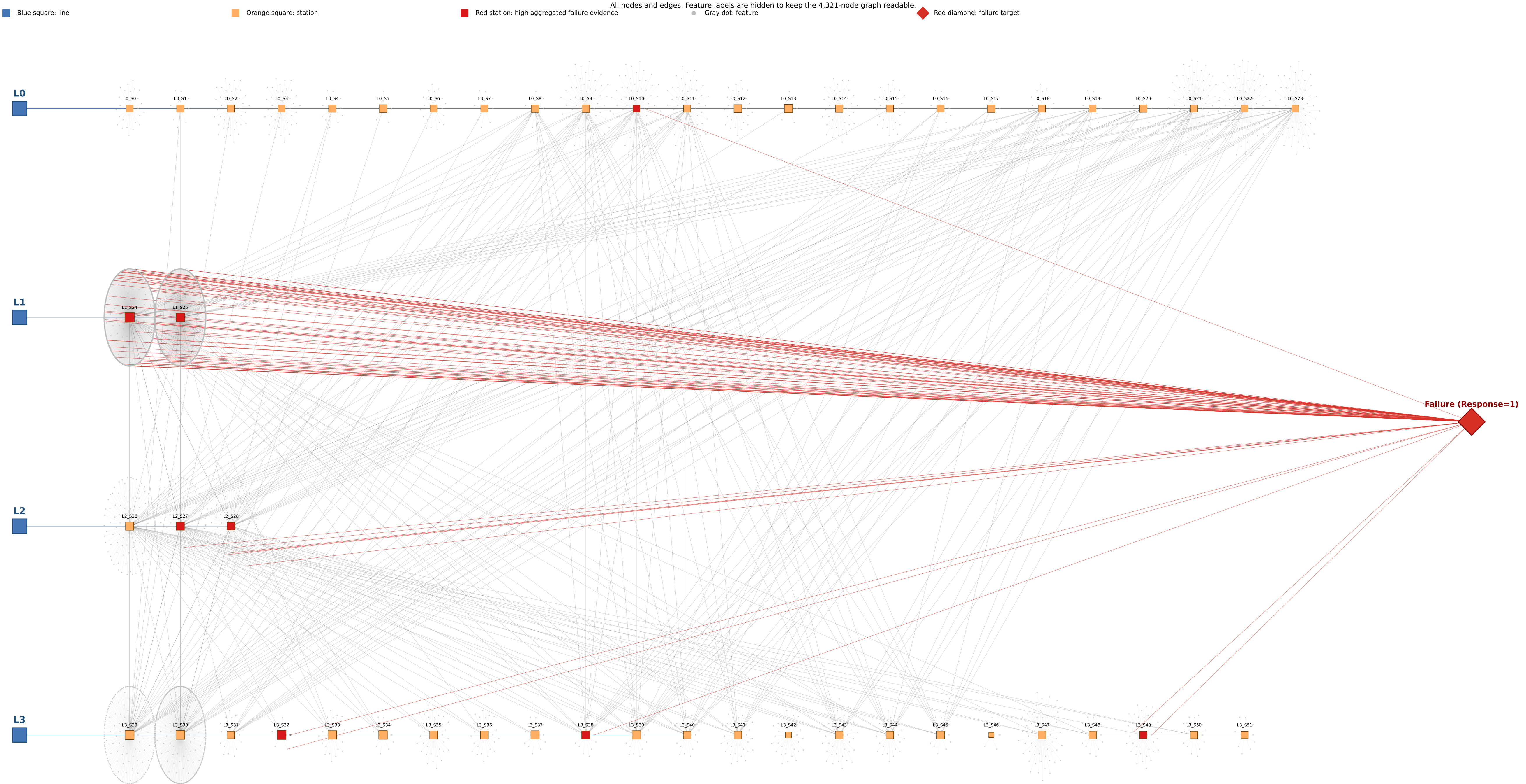
label	line	station	critical_score	failure_evidence
L3_S32_F3850	L3	L3_S32	100.00	0.067988
L2_S27_F3144	L2	L2_S27	34.30	0.023317
L1_S24_F1844	L1	L1_S24	30.73	0.020889
L1_S24_F1846	L1	L1_S24	26.25	0.017843
L1_S24_F1723	L1	L1_S24	20.29	0.013797
L1_S24_F1778	L1	L1_S24	18.49	0.012571
L1_S24_F1667	L1	L1_S24	17.18	0.011678
L1_S24_F1498	L1	L1_S24	15.31	0.010412
L1_S24_F1713	L1	L1_S24	13.27	0.0090237
L1_S24_F1842	L1	L1_S24	12.42	0.0084438
L1_S24_F1758	L1	L1_S24	12.25	0.0083253
L1_S24_F958	L1	L1_S24	11.82	0.0080392
L1_S24_F1672	L1	L1_S24	11.05	0.0075121
L1_S24_F1609	L1	L1_S24	10.90	0.0074093
L1_S24_F1122	L1	L1_S24	10.75	0.0073106
L3_S49_F4211	L3	L3_S49	9.96	0.0067721
L1_S24_F1490	L1	L1_S24	9.95	0.0067678
L1_S24_F1002	L1	L1_S24	9.92	0.0067471
L1_S24_F1176	L1	L1_S24	9.63	0.0065496
L1_S25_F2990	L1	L1_S25	8.46	0.0057489

Graph pages included after this summary

- Full manufacturing knowledge graph overview: all nodes and edges, with feature labels hidden.
- Station-level flow: line nodes, stations, manufacturing transitions, and highest failure associations.
- Line detail pages: each line with its stations and the top critical feature per station.

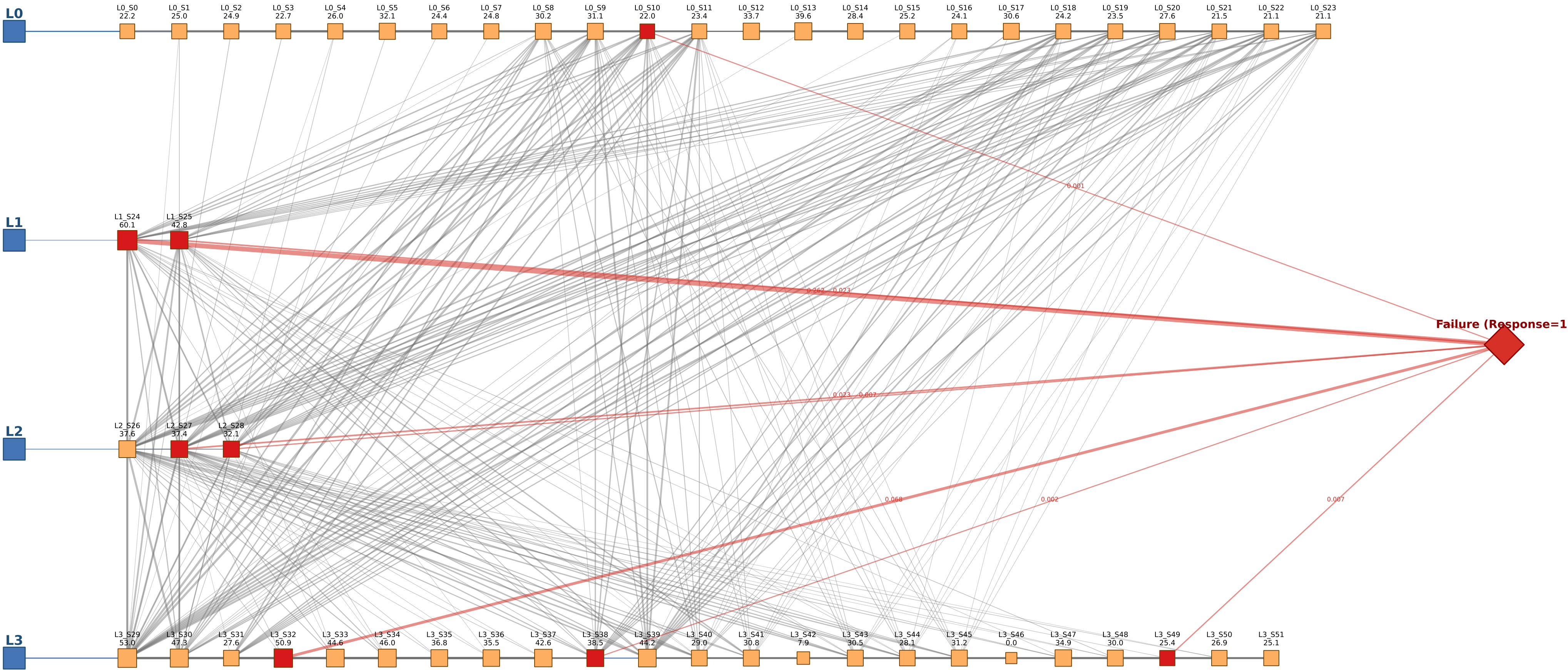
Use a desktop PDF viewer and zoom in on graph pages for detailed inspection.

Full manufacturing knowledge graph overview



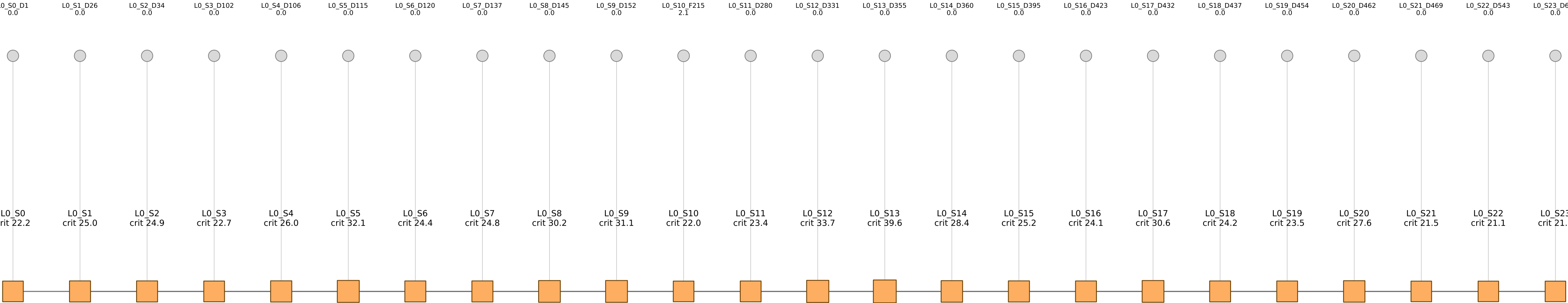
Station-level flow and failure association

Station label includes critical score. Red stations have the highest aggregated failure evidence.



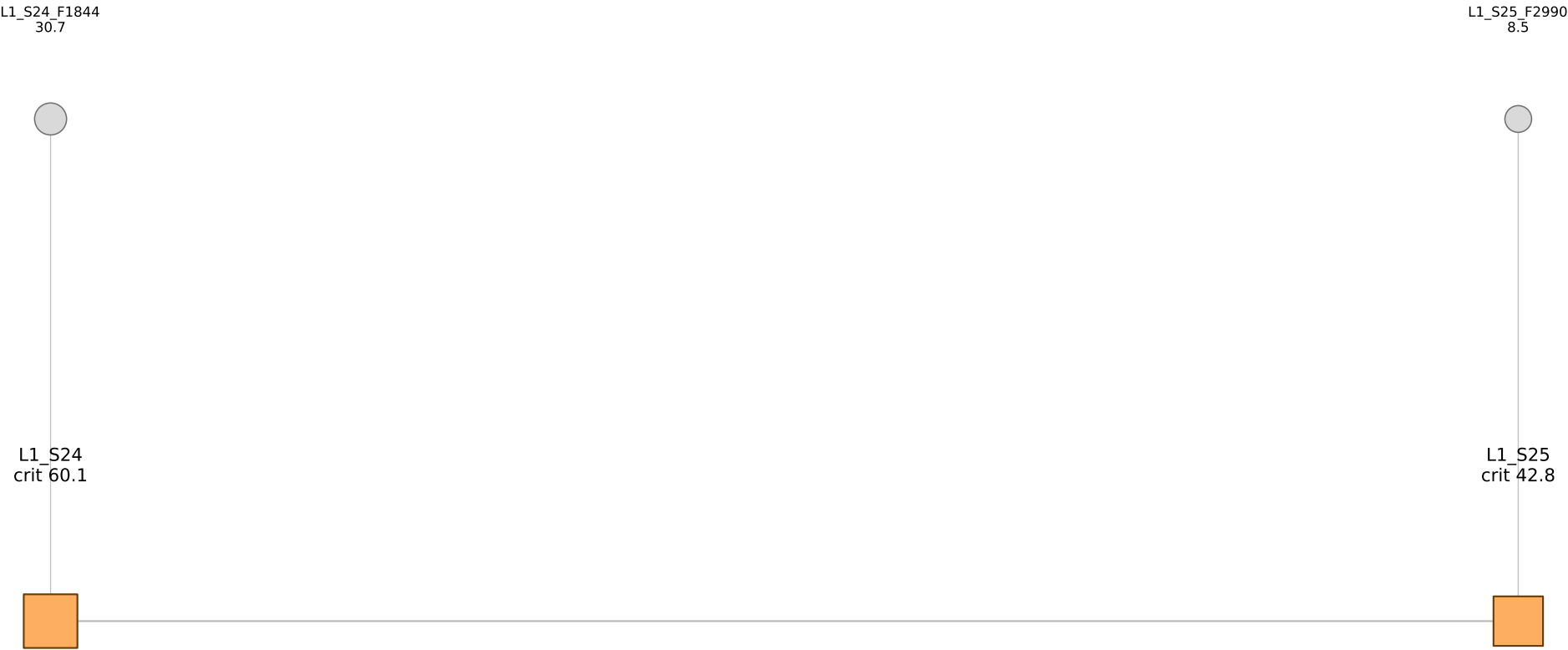
L0 detail: stations and top critical features

Each station shows its top critical feature. Feature nodes are labeled with critical score.



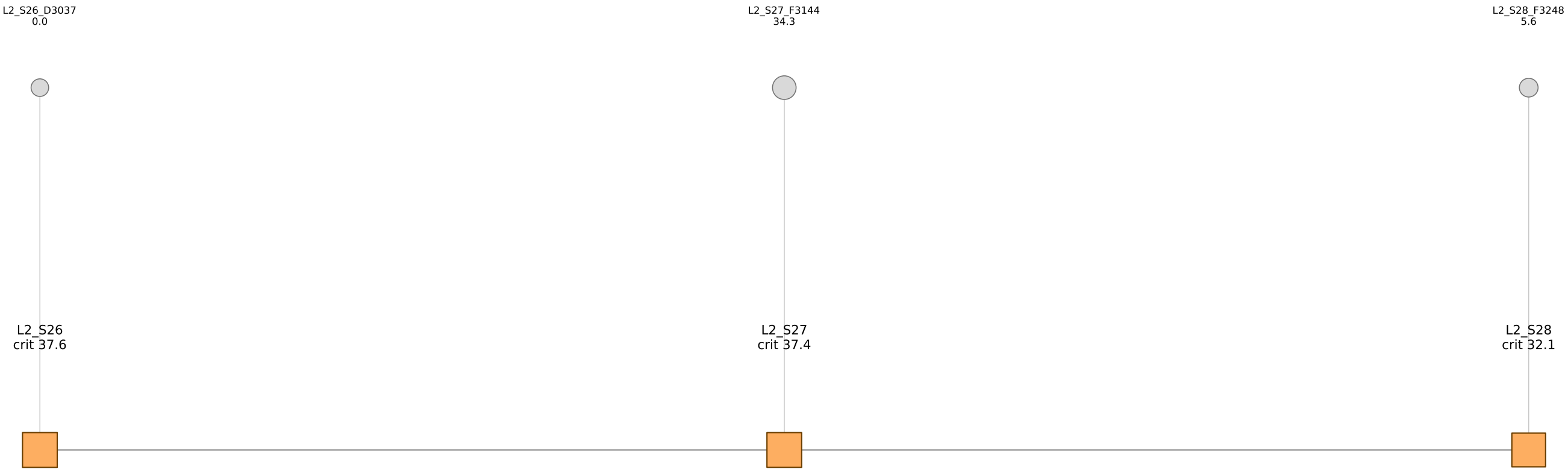
L1 detail: stations and top critical features

Each station shows its top critical feature. Feature nodes are labeled with critical score.



L2 detail: stations and top critical features

Each station shows its top critical feature. Feature nodes are labeled with critical score.



L3 detail: stations and top critical features

Each station shows its top critical feature. Feature nodes are labeled with critical score.

